

Economic Optimization of Insurances using Active Learning

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Part 1: Economically Optimized Insurance Fraud Detection with Active Learning

Objectives



- **Minimize fraudulent claims** which lead to direct financial losses to the lender.
- **Capture data distribution drift** leveraging new data streams and historical biases.
- **Maximize ROI on labeling**, i.e. best resource expenditure to maximize learnability.
- Maintain **lender reputation** and customer trust.

Data Sourcing

Due to privacy concerns, we statistically analyzed and adopted internal schema to synthetically generate 48 features (mix of continuous and categorical ones).

Generate the fraud rules used for naive flagging to:

- Help learn patterns from bias data & test on unbiased set.
- Simulate a financial corporation's internal process of not evaluating each point by models directly.

1) INPUT: Dataset

Let the input be a dataset $D = \{x_i\}_{i=1}^n$, where each row x_i contains the following features:

Symbol	Feature (Column Name)	Description
s_i	sum_insured	Total sum insured for the policy
p_i	premium_with_tax	Premium including tax
b_i	previous_insurance_buildings	Number of previously insured buildings
a_i	accidental_damage_premium	Accidental damage premium amount
u_i	unoccupied_days	Number of days the property is unoccupied
c_i	no_of_claims_5yr	Number of claims in the last 5 years

2) PROCESSING

For each record x_i , compute the claim ratio: $r_i = \frac{s_i}{p_i + 1}$

A fraud flag is raised if **ANY** of the following conditions is true:

Fraud Rules

- $r_i > 1500$ AND $b_i = 0$ AND $c_i > 0$
Very high claim ratio, no prior insured buildings, at least one claim
- $a_i > 0$ AND $u_i \geq 30$ AND $c_i > 0$
Has accidental damage premium, property unoccupied for ≥ 30 days, at least one claim
- $c_i \geq 3$
Three or more claims in the last 5 years

3) OUTPUT

The function returns a boolean fraud flag vector $f = (f_1, f_2, \dots, f_n)$, $f_i \in \{0, 1\}$ where for each record x_i ,

$$f_i = 1 \left[\left(\frac{s_i}{p_i + 1} > 1500 \wedge b_i = 0 \wedge c_i > 0 \right) \vee (a_i > 0 \wedge u_i \geq 30 \wedge c_i > 0) \vee (c_i \geq 3) \right]$$

$1[\cdot]$ is the indicator function, which equals 1 if the condition is true, 0 otherwise.



In words: A claim is flagged as fraudulent if it satisfies any one of the above three rule-based conditions.

Data Sourcing

STEP 0: SENTINEL -1 HANDLING

For each feature j in the sentinel list S :

$$m_j(x_i) = 1 [x_{ij} = -1]$$

$$\tilde{x}_{ij} = \begin{cases} x_{ij}, & x_{ij} \neq -1, \\ \text{median}\{x_{ej} : x_{ej} \neq -1\}, & x_{ij} = -1. \end{cases}$$

$m_j(x_i)$: Missing indicator (1 if sentinel -1, else 0)

$\tilde{x}_{ij}$: Sentinel-cleaned value used for all computations

VARIABLES AND INDICATORS

Symbol	Meaning (original feature)
$\tilde{s}_i$	sum_insured (clean)
$\tilde{r}_i$	reinstatement_value (clean)
d_i	claim_days_differenc
a_i	claim_loss_date_to_policy_str_date_delta
g_i	claim_loss_datet_to_policy_ply_icp_dt_delta
b_{1i}, b_{2i}, b_{3i}	risk_band1, risk_band2, risk_band3
o_i	oc_code (1 if present)
t_i	claim_loss_type_th (1 if present)

Missing Indicators (m_j)

$m_{\text{credit}}(x_i)$	Missing indicator for credit_risk_score
$m_{\text{floor}}(x_i)$	Missing indicator for floor_space
$m_{\text{ridge}}(x_i)$	Missing indicator for ridge_height

$1[\cdot]$: Indicator function (1 if condition true, else 0)

Let x_i be row i of the dataset, $i = 1, \dots, n$. n = number of rows. t = target_fraud_rate. $q_p(\cdot)$ = p -th percentile.

A. OPPORTUNISTIC FRAUD SCORE $S_A(i)$

$$1 \left[\frac{\tilde{s}_i}{\max(\tilde{r}_i, 5000)} > 1.3 \right] \\ \left(e^{2.5 \left(\frac{\tilde{s}_i}{\max(\tilde{r}_i, 5000)} - 1.3 \right)} - 1 \right) \\ + 0.9 \mathbf{1} [\tilde{s}_i \equiv 0 \pmod{1000}] \\ + 0.4 \mathbf{1} [\tilde{s}_i \equiv 0 \pmod{500}, \tilde{s}_i \not\equiv 0 \pmod{1000}] \\ + 0.6 m_{\text{credit}}(x_i) \\ + 4.0 \mathbf{1} \left[\frac{\tilde{s}_i}{\max(\tilde{r}_i, 5000)} > 1.3 \right] m_{\text{credit}}(x_i) \mathbf{1} [\tilde{s}_i \equiv 0 \pmod{1000}]$$

B. PREMEDITATED FRAUD SCORE $S_B(i)$

$$2.8 e^{-d_i^2/18} \\ + 1.5 e^{-(d_i - 120)^2/600} \\ + 3.5 \mathbf{1} [a_i < 60] \mathbf{1} [d_i < 5] \\ + 2.5 \mathbf{1} [g_i < 5]$$

C. ORGANIZED RING FRAUD SCORE $S_C(i)$

$$5 \left(\frac{\max(0, b_{1i})}{30} \right) \left(\frac{\max(0, b_{2i})}{30} \right) \left(\frac{\max(0, b_{3i})}{30} \right) \\ + 2.8 \mathbf{1} [o_i = 1] \\ + 2.2 m_{\text{floor}}(x_i) m_{\text{ridge}}(x_i) \\ + 1.8 \mathbf{1} [t_i = 1] \mathbf{1} [b_{1i} > 20]$$

INTERACTION I1: CROSS-ARCHETYPE SYNERGY $P(i)$

$$P(i) = 1.5 \left(\frac{\min(S_A(i), 3)}{3} \right) \left(\frac{\min(S_B(i), 3)}{3} \right) \left(\frac{\min(S_C(i), 3)}{3} \right)$$

INTERACTION I2: SHADOW PROFILE (XOR-STYLE) $H(i)$

$$H(i) = 2.5 \mathbf{1} [\tilde{s}_i > q_{0.8}(\tilde{s})] \mathbf{1} \left[\bigcap_{j \in \{\text{sum_insured, reinstatement_value, credit_risk_score}\}} m_j(x_i) \geq 2 \right]$$

RAW FRAUD SCORE

$$R_i = S_A(i) + S_B(i) + S_C(i) + P(i) + H(i) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, 0.15^2)$$

ROBUST PROBABILITY SCORE (MIN-MAX USING PERCENTILES)

$$\hat{p}_i = \text{clip} \left(\frac{R_i - q_{0.01}(R)}{q_{0.995}(R) - q_{0.01}(R) + 10^{-9}}, 0, 0.999 \right)$$

FRAUD LABEL (TOP- t FRACTION)

$$y_i = \mathbf{1} [\hat{p}_i \geq q_{1-t}(\hat{p})], \quad t = \text{target_fraud_rate}. \quad y_i \in \{0, 1\}$$

OUTPUT

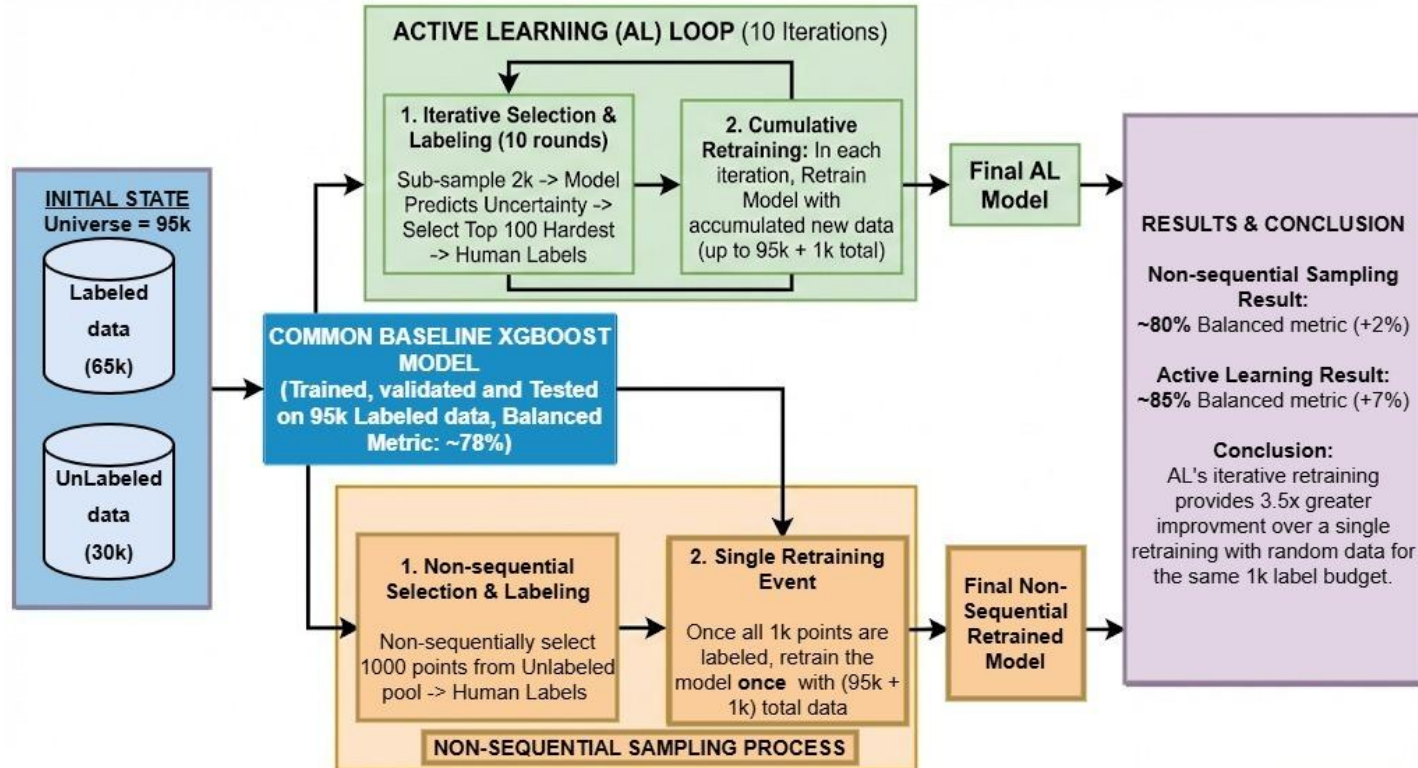
- Fraud probability score $\hat{p}_i \in [0, 0.999]$ for each row i .
- Fraud label $y_i \in \{0, 1\}$ such that approximately a fraction t are labeled 1.

The function returns the input dataset with two new columns: **fraud_probability_score** ($\hat{p}_i$) and **fraud_label** (y_i).

Notes: (1) All computations use sentinel-cleaned values $\tilde{x}_{ij}$; missing indicators $m_j(x_i)$ are fraud signals.

(2) Percentiles are computed over all rows. (3) $\text{clip}(a, 0, 0.999)$ restricts values to $[0, 0.999]$.

Methodology



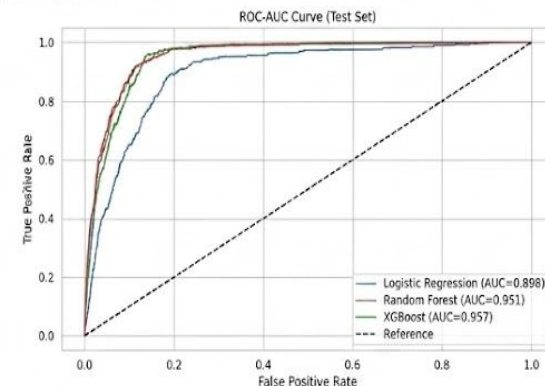
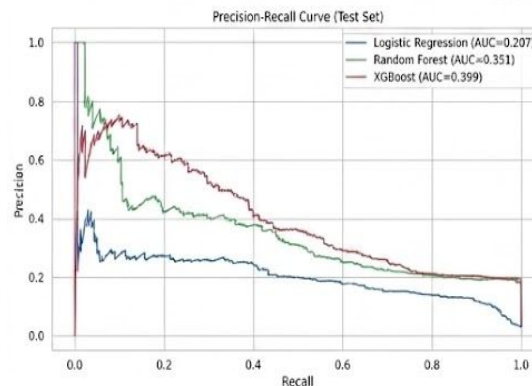
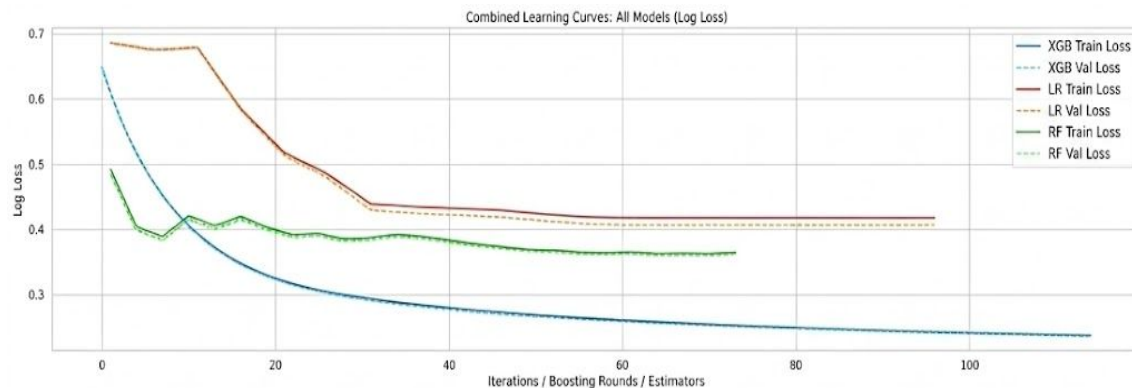
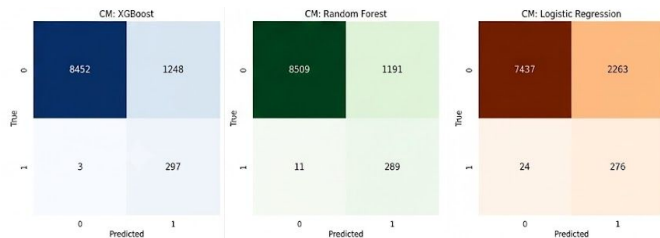
Results



Accuracy based models (One-shot Baselines):

Train (45k) and val (20k) from labeled, while Test (10k) from unlabeled set.

	Model	ROC-AUC	F1-Score	Precision	Recall
0	Logistic Regression	0.897624	0.194435	0.108704	0.920000
1	Random Forest	0.951370	0.324719	0.195270	0.963333
2	XGBoost	0.956589	0.321951	0.192233	0.990000

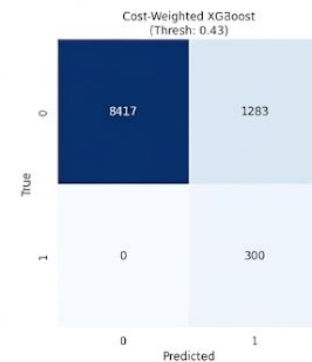
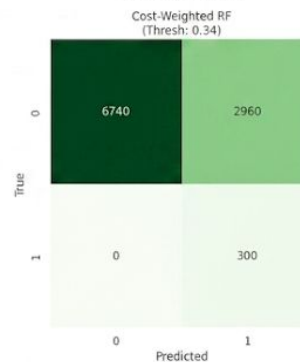
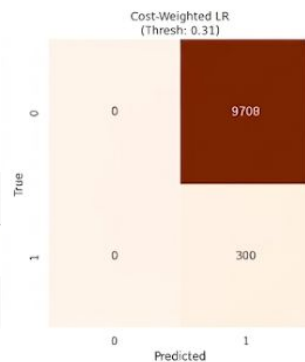
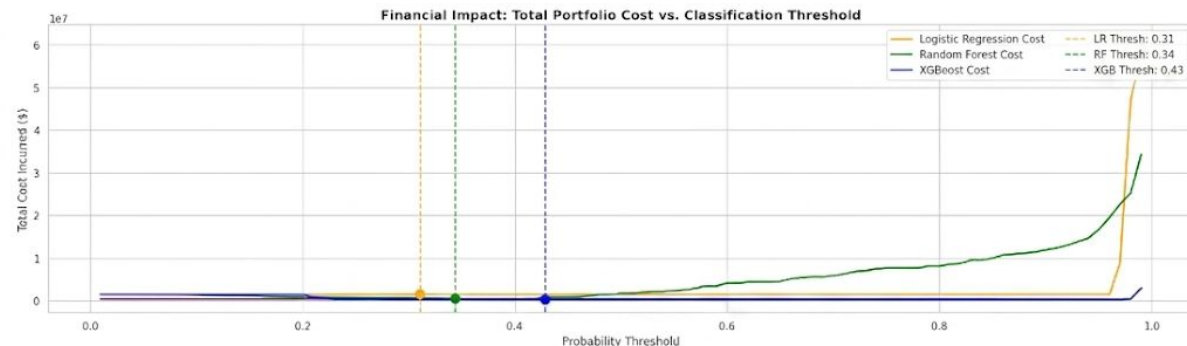


Results



Business cost based models (One-shot Baselines):

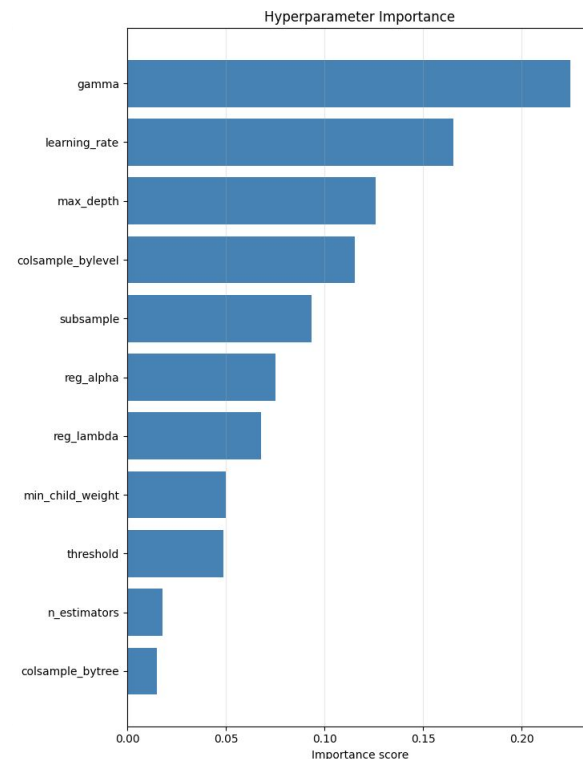
Train (45k) and val (20k) from labeled, while Test (10k) from unlabeled set.



	Model	Opt_Threshold	Cost / Row	Total Cost (Test)	Precision	Recall	F1-Score	ROC-AUC
0	Logistic Regression	0.310748	148.500	1485000.0	0.030000	1.0	0.058252	0.506235
1	Random Forest	0.343953	47.400	474000.0	0.092025	1.0	0.168539	0.943663
2	XGBoost	0.428643	22.245	222450.0	0.189514	1.0	0.318640	0.954396

Tuning & Parameter Importance Analysis:

Outcome	Condition	Cost	Interpretation
True Positive (TP)	Predicted Fraud = 1, Actual Fraud = 1	£100	Cost incurred for investigating a correctly identified fraudulent claim.
False Positive (FP)	Predicted Fraud = 1, Actual Fraud = 0	£150	Unnecessary investigations, customer inconvenience, and reputational impact.
True Negative (TN)	Predicted Fraud = 0, Actual Fraud = 0	£0	No action required; therefore, no associated cost.
False Negative (FN)	Predicted Fraud = 0, Actual Fraud = 1	$0.9 \times \text{Sum Insured}$	Cost of an undetected fraudulent claim payout = 90% of the insured amount.

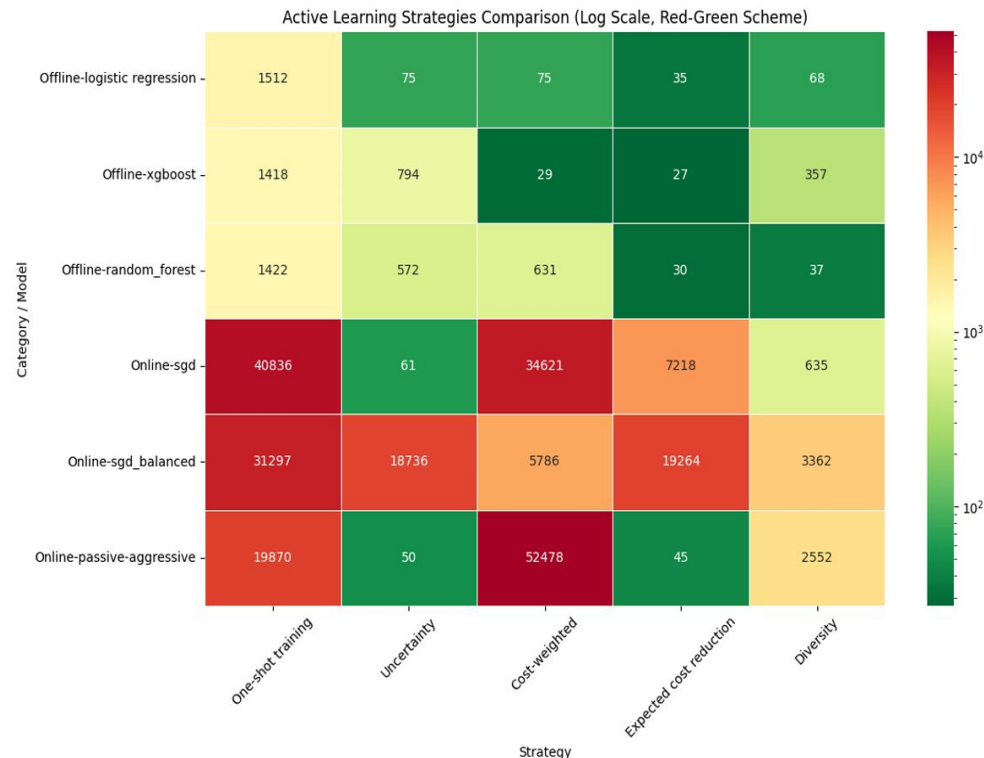


Results



Online vs Offline Active Learning Setup

- **Cost-Weighted:** selects highly uncertain predictions & scales them by their financial impact (insured amount).
- **Uncertainty:** Selects the most ambiguous points closest to the 50% decision boundary, where the model is least confident.
- **Diversity (Cluster + Uncertainty):** Combines model's uncertainty with spatial distances from K-means clusters to avoid querying redundant data points.
- **Expected Cost Reduction:** Selects samples with the highest potential financial regret by weighing the monetary penalties of making a false positive versus a false negative.



Results

Hybrid AL Strategy

Intersecting all basic strategies, we are essentially creating a "consensus filter". The model will now query a row if it is mathematically uncertain, financially impactful, structurally diverse, and reduces expected misclassification cost.

HYBRID AL vs AL (Cost-Weighted Uncertainty)

Δ Hybrid AL vs AL (positive = Hybrid AL better for AUC/recall/f1)

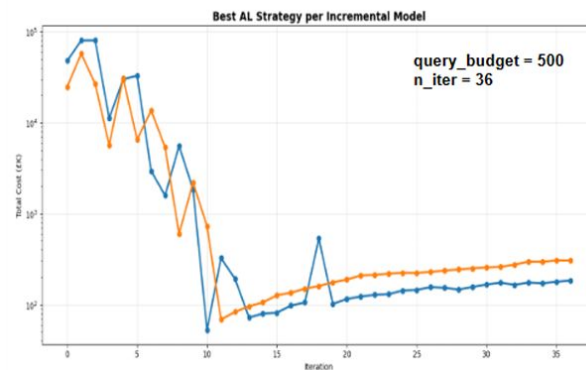
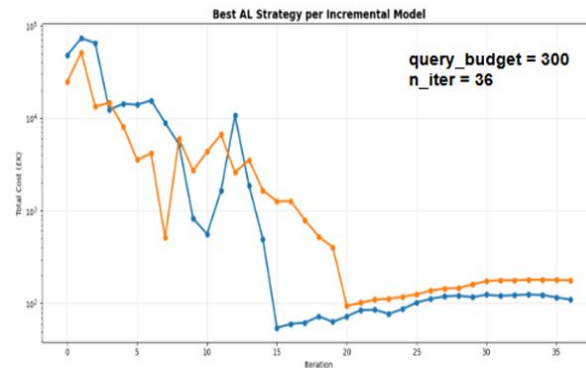
AUC	: +0.0016
Business cost	: -1,350.00 ✔ cheaper
Recall	: +0.0000
F1	: +0.0006

df_unlabeled total rows	: 30,000
seen rows (union, deduped)	: 2,313

model	Baseline (labeled pool only)	Non-Sequential (random 1k unlabeled)	Active Learning (Cost-weighted Uncertainty)	Hybrid AL (Multi-Strategy)
auc	0.9553	0.9561	0.9547	0.9563
biz_cost	617400.0000	616800.0000	609300.0000	607950.0000
cost_per_row	22.2832	22.2615	21.9908	21.9421
tp	843.0000	843.0000	843.0000	843.0000
fp	3554.0000	3550.0000	3500.0000	3491.0000
tn	23310.0000	23314.0000	23364.0000	23373.0000
fn	0.0000	0.0000	0.0000	0.0000
precision	0.1917	0.1919	0.1941	0.1945
recall	1.0000	1.0000	1.0000	1.0000
f1	0.3218	0.3220	0.3251	0.3257

Conclusion

- The model achieves **information saturation at $\approx 12\%$ of the dataset**. Having more iterations results in diminishing returns and rising FP costs.
- **Online Active Learning** gave promising results, helping not only to minimize losses against frauds, but also allowing iterative model training with only the **few new data-points** that comes in, preventing obsolete patterns from old dataset to hamper learning from newer ones.
- Hybrid AL, i.e. selecting points by intersection of various strategies, help reduce cost further and beat the naive ones.





Part 2: Economically Optimized Pure Payout Pricing via Stratified Expected Deviance Reduction (SEDR) in Active Learning

Objectives



- **Maximize the labeling ROI** and **predictive pricing accuracy** of a single **Tweedie** Gradient Boosted Decision Tree (GBDT) model predicting pure insurance payouts.
- **Decouple** the epistemic model disagreement from aleatoric claim severity, thereby **prevent a systemic collapse** in frequency calibration (Zero-Mass Calibration Error).
- Evaluate the **model's economic utility** by tracking ZMCE to maintain distinguishability between claim data-points.
- Preserve economic rank-ordering utility (Gini coefficient) while accelerating convergence to the true pure payout (Tweedie Deviance).

Tweedie

- Member of the Exponential Dispersion Model (EDM) family.
- Parameterized by the mean μ , the dispersion ϕ , and the variance power p .
- $p \in (1, 2) \rightarrow$ physically correct regime for us.
- Tweedie deviance is the natural loss function for this family & is the difference between the log-likelihoods of the saturated & the fitted model.
- Aggregate out-of-sample deviance, weighted by policy exposure w_i , is the primary metric .

Power Parameter (p)	Distribution	Variance Function	Characteristics
$p = 0$	Normal	1	Constant variance, symmetric
$p = 1$	Poisson	μ	Variance equals mean; count data
$1 < p < 2$	Compound Poisson-Gamma	μ^p	Zero-inflated with positive continuous mass
$p = 2$	Gamma	μ^2	Strictly positive, continuous
$p = 3$	Inverse Gaussian	μ^3	Heavy right tail

$$\mathbb{E}[Y] = \mu$$

$$\text{Var}(Y) = \phi \mu^p, \quad 1 < p < 2$$

Data Sourcing

French Motor Third-Party Liability (MTPL) dataset.

Total Policies: 678k+

Exact Zeros: 96.32%

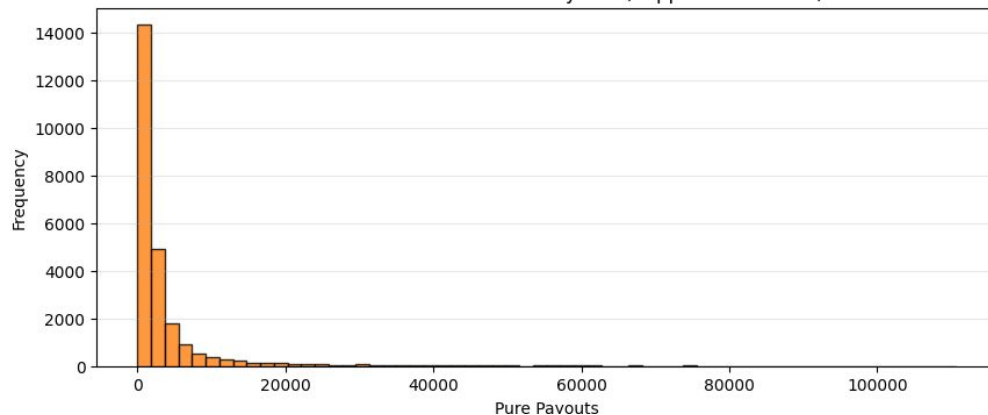
$$Y_i = \frac{\text{TotalClaimAmount}_i}{\text{Exposure}_i}$$

Exposure is a structural component of the Tweedie likelihood. In a properly formulated Tweedie GLM, the implied Poisson rate must scale linearly with exposure.

Severity parameters must remain independent of exposure so that all evaluation metrics involving the zero-mass probability must explicitly account for each policy's exposure weight.

Failure leads to **Exposure Bias Error**, ie, shorter-duration policies are incorrectly evaluated as if they represent full-year risks.

Distribution of Positive Pure Payouts (Clipped at 99th Pct)



Feature	Description
Area	City density code based on French administrative classification
VehPower	Ordered categorical variable for vehicle engine power
VehAge	Vehicle age in years
DrivAge	Driver age in years (legal minimum: 18 in France)
BonusMalus	Insurance modifier in [50, 350]; values < 100 indicate safe driving history
VehBrand	Encoded vehicle manufacturer category
VehGas	Fuel type (regular or diesel)
Density	Number of inhabitants per km ² in the policyholder's municipality
Region	French administrative region code
Exposure	Policy duration in years (a continuous risk weight in (0,1])

Metrics

- Exposure weighted **Tweedie Deviance** (Pricing Accuracy):

It critically needs estimation of the dispersion parameter ϕ from the training set to prevent induced deflation of ZMCE due to oracle leakage.

$$d(y, \hat{\mu}) = 2 \left(\frac{\max(y, 0)^{2-p}}{(1-p)(2-p)} - \frac{y\hat{\mu}^{1-p}}{1-p} + \frac{\hat{\mu}^{2-p}}{2-p} \right)$$

$$D(\mathbf{y}, \hat{\boldsymbol{\mu}}) = \frac{1}{\sum_i w_i} \sum_i w_i \cdot d(y_i, \hat{\mu}_i)$$

$$\hat{\phi} = \frac{1}{n_{train} - q} \sum_{i \in \mathcal{X}_{train}} \frac{(y_i - \hat{\mu}_i)^2}{\hat{\mu}_i^p}$$

- Zero-Mass Calibration Error (**ZMCE**): absolute difference between the empirically observed proportion of zero claims in the test set and the model's implied probability of a zero claim.

$$\text{ZMCE} = \left| \frac{1}{N_{test}} \sum_{i \in \mathcal{X}_{test}} \mathbf{1}[y_i = 0] - \frac{1}{N_{test}} \sum_{i \in \mathcal{X}_{test}} e^{-\lambda_i} \right|$$

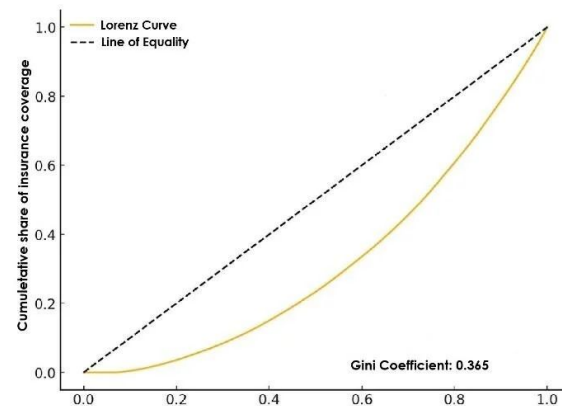
$$\lambda_i = \frac{w_i \cdot \hat{\mu}_i^{2-p}}{[\hat{\phi}(2-p)]}$$

Metrics

- **Normalized Gini Index** (Economic Rank-Ordering): ability of a pricing model to discriminate between profitable and loss-making risks. It is coupled with Lorenz curve, which is constructed as follows:

1. Sort policyholders in ascending order of predicted pure payout μ_i .
2. Compute the cumulative fraction of total exposure (X-axis) & cumulative fraction of total realized claims (Y-axis).

$$G = 1 - 2 \int_0^1 L(x), dx \approx 1 - 2 \cdot \text{AUC}(\text{Lorenz Curve})$$



- **Total Financial Bias:** quantifies the aggregate systemic under-reserving or over-reserving risk.

$$\text{Bias} = \frac{\sum_i \hat{y}_i - \sum_i y_i}{\sum_i y_i}$$

- **Spearman Rank Correlation:** non-parametric measure of the monotonic alignment between predicted risk ranks and actual claim costs. Unlike Pearson correlation, ρ is robust to the extreme skewness of the claim distribution.

$$\rho = 1 - \frac{6 \sum d_i^2}{N(N^2 - 1)}$$

- **Label Efficiency AUC:** area under the deviance reduction curve across all the active learning iterations, normalized against the terminal deviance of the single-shot baseline.

$$\text{Label Efficiency} = 1 - \frac{\text{AUC}(\text{AL Deviance Trajectory})}{\text{AUC}(\text{Baseline Plateau})}$$

Metrics - Traps

MAE trap:

- Global Mean Absolute Error optimizes for the **median** of the distribution.
- $\geq 90\%$ of claims are zero, so global MAE forces to predict $\sim \$0$ for almost every policyholder, creating the mathematical illusion of low error while chronically underpricing the portfolio.

Naive Variance achieves a superficially low global MAE by underpricing the zero-claim majority, while the SEDR architecture achieves strictly lower MAE across every single non-zero claim tier using:

Stratified Exposure-Weighted MAE (WMAE)

ZMCE trap:

A low ZMCE confirms that the model's internal decomposition of frequency and severity is correctly calibrated.

But, a ZMCE near zero that coincides with a high deviance and low Gini index, is the mathematical signature of calibration collapse, ie, the model is predicting $P(Y = 0) \approx 1$ for the entire population by driving $\lambda \rightarrow 0$, rather than genuinely learning the frequency structure.

Tier	Definition
Zero Claims	$y_i = 0$
Low Claims	$0 < y_i \leq P_{50}$ of non-zero claims
Medium Claims	$P_{50} < y_i \leq P_{75}$
High Claims	$P_{75} < y_i \leq P_{95}$
Catastrophic Tail	$y_i > P_{95}$

Methodology

- **One-shot baseline:** full terminal data budget (seed + all n iterations $\times$ query size).
- **Random (Naive Sequential Learning):** selects the next batch **uniformly at random** from X_{pool} , preserving the correct proportion of **zero to non-zero** value.
- **Naive Variance AL:** ranks the unlabeled pool by **raw predicted mean** $\hat{\mu}$ (direct proxy for Tweedie variance), and queries the top- B instances at each iteration. This is the precise mechanism by which **calibration collapse** is induced.
- **Epistemic-Only AL:** queries purely by epistemic model disagreement (as measured by **single-model tree variance**) without stratified frequency anchor. By hunting pure model confusion without strata constraints, the acquisition function fixates on the noisy decision boundary between safe drivers & achieves good calibration & low deviance but **destroys Gini** by oversampling safe drivers near model noise boundaries, leaving the model with **no ability to rank-order genuine high-risk profiles**.

Methodology

Stratified Expected Deviance Reduction (SEDR):

- **Stratified Risk Partitioning:** unlabeled pool $\mathcal{X}_{pool}$ is partitioned into 5 risk quintiles based on the current model's predicted mean (Q_1: safest to Q_5: riskiest).

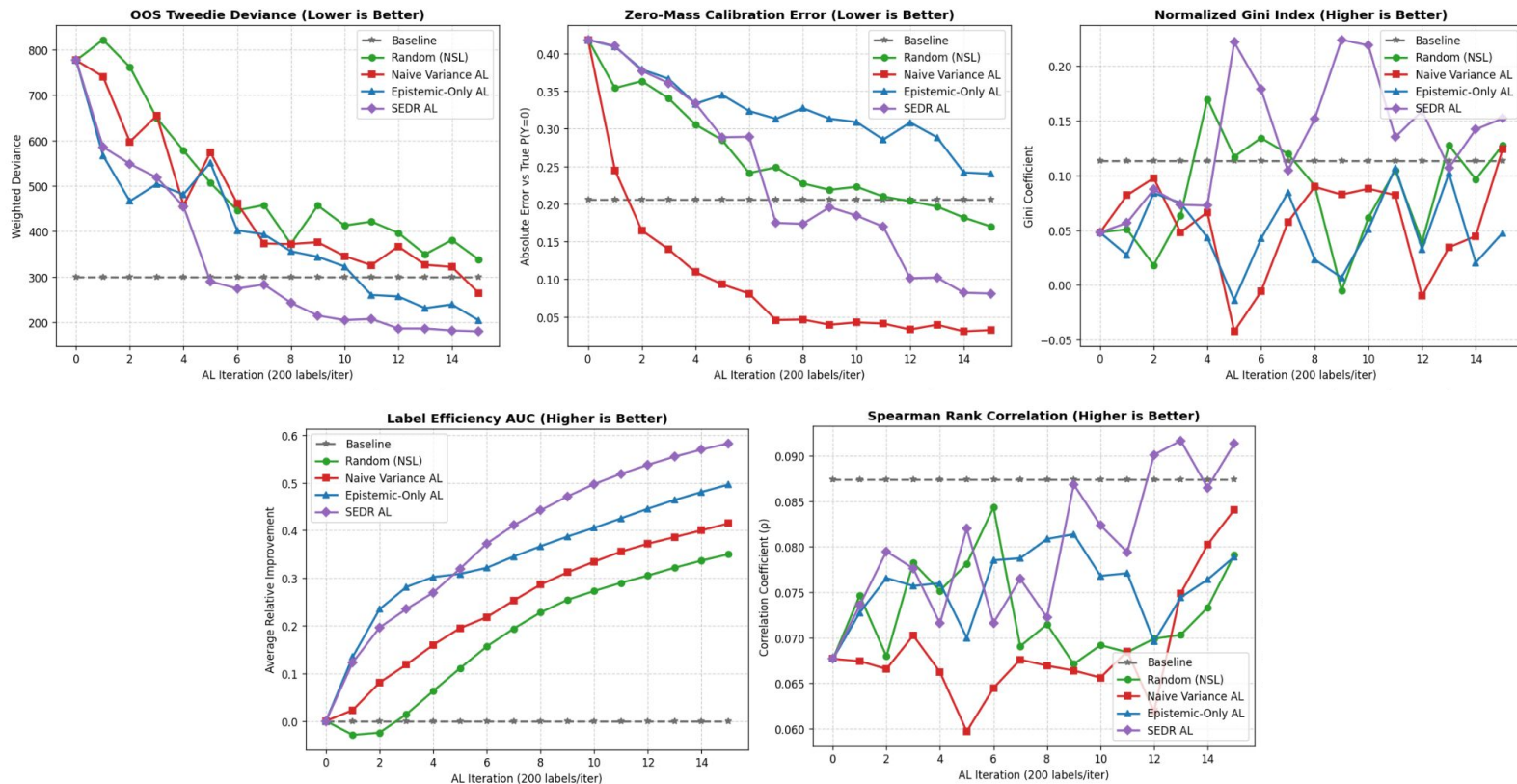
$$Q_k = \left\{ x_i \in \mathcal{X}_{pool} : \hat{\mu}_i \in \left[F^{-1} \left(\frac{k-1}{5} \right), F^{-1} \left(\frac{k}{5} \right) \right) \right\}, \quad k = 1, \dots, 5$$

where $F^{-1}(\cdot)$ is the empirical quantile function of $\hat{\mu}$ over $\mathcal{X}_{pool}$.

- **Query Budget Allocation:** 30% allocation to Q_1 acts as a Poisson Frequency Anchor, preventing the dispersion parameter ϕ from inflating uncontrollably.
- **Epistemic Acquisition:** Within each non-zero stratum Q_k ($k \geq 2$), instances are ranked by their epistemic uncertainty score & top- b Q_k instances are selected for next round.

$$b_{Q_1} = \lfloor 0.30 \times B \rfloor, \quad b_{Q_{2:5}} = \left\lfloor \frac{0.70 \times B}{4} \right\rfloor \text{ per stratum}$$

Results

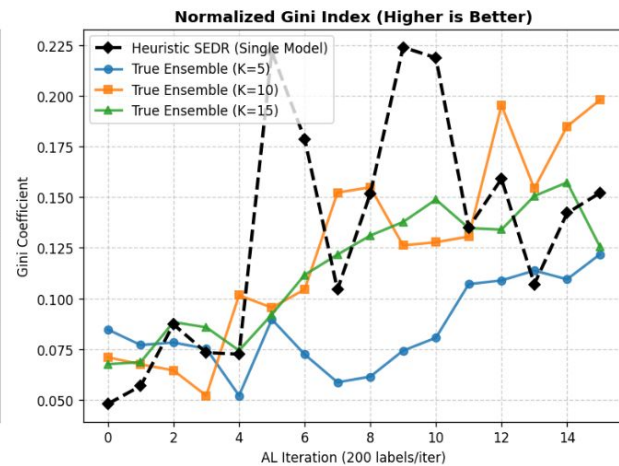
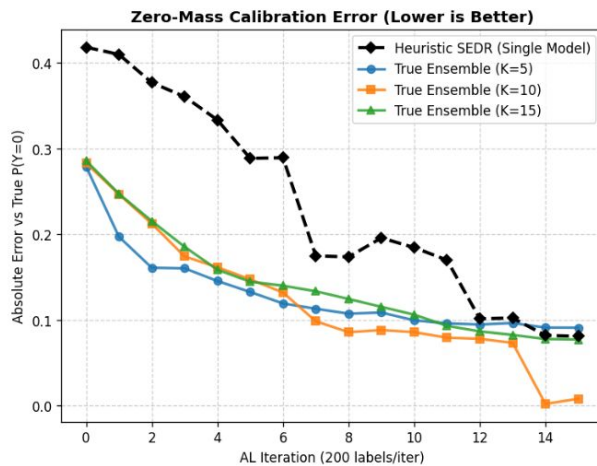
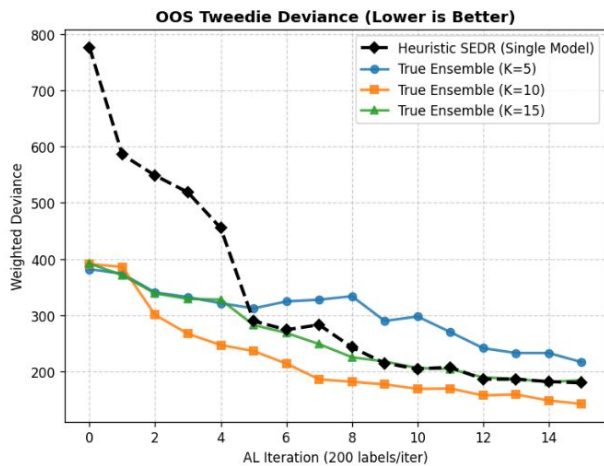


Results

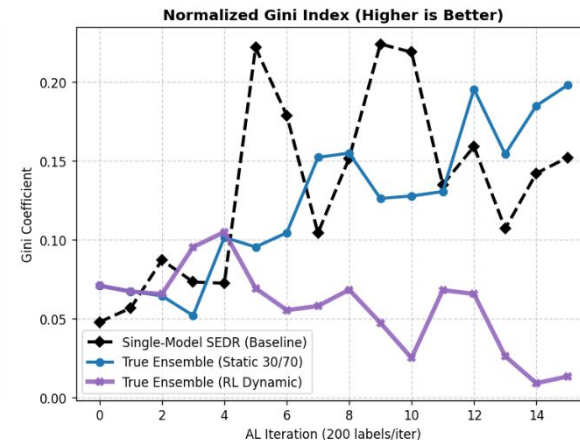
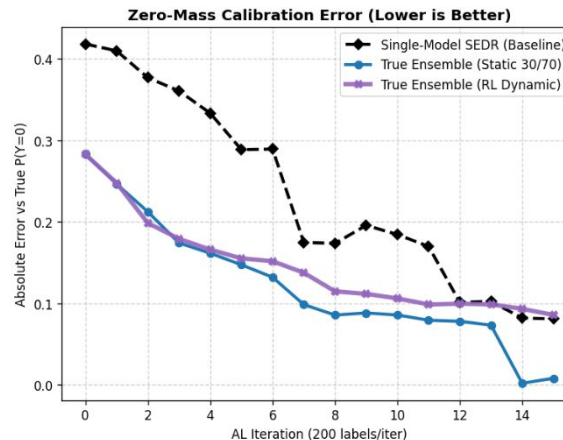
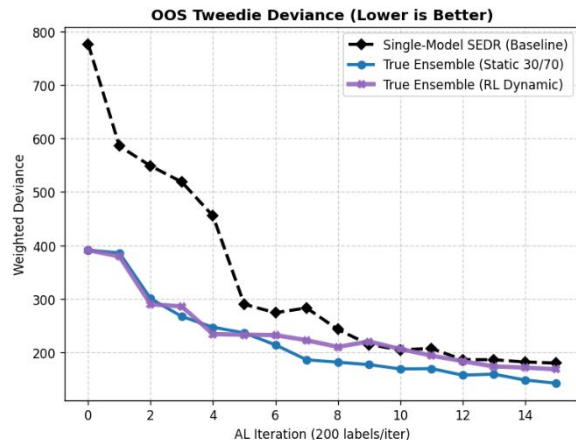
	Claim Tier	Policy Count	Exposure Weight	Baseline (Single-Shot)	Random (NSL)	Naive Variance AL	Epistemic-Only AL	SEDR AL (Optimal)
0	1. Zero Claims (\$0)	9638	4979.5	12.13	10.62	7.40	11.77	27.28
1	2. Low Claims (>\$0 to 1172)	130	110.6	681.88	686.41	683.74	684.65	669.46
2	3. Medium Claims (1172 to 2548)	109	82.7	1604.40	1606.73	1610.60	1604.69	1560.69
3	4. High Claims (2548 to 23420)	104	44.5	5753.32	5759.46	5759.08	5749.04	5722.47
4	5. Catastrophic Tail (>23420)	19	3.3	73440.70	73439.85	73441.52	73440.76	73422.63
5	6. OVERALL PORTFOLIO	10000	5220.6	147.37	146.12	143.05	147.05	160.59

- Zero Claims:** SEDR charges an average of \$27.28, while Naive Variance charges \$7.40. Naive Variance appears to "win" by \$20, but this gap represents a ~\$192,000 aggregate **under-reservation on a portfolio** of 9,638 zero-claim policies.
- Low Claims:** SEDR MAE (669.46) < Naive Variance MAE (683.74).
- Medium Claims:** SEDR (1560.69) < Naive Variance (1610.60).
- High Claims:** SEDR (5722.47) < Naive Variance (5759.08).
- Catastrophic Tail:** SEDR (73422.63) < Naive Variance (73441.52)
- Overall Portfolio** - MAE trap evident! → SEDR (160.59) > Naive Variance (143.05)

Results



Results



	Claim Tier	Policy Count	Exposure Weight	Single-Model SEDR (Baseline)	True Ensemble (Static 30/70)	RL-Driven Dynamic SEDR
0	1. Zero Claims (\$0)	9638	4979.5	27.28	29.49	26.23
1	2. Low Claims (>\$0 to 1172)	130	110.6	669.46	662.49	666.62
2	3. Medium Claims (1172 to 2548)	109	82.7	1560.69	1578.19	1584.70
3	4. High Claims (2548 to 23420)	104	44.5	5722.47	5715.99	5734.10
4	5. Catastrophic Tail (>23420)	19	3.3	73422.63	73403.47	73417.15
5	6. OVERALL PORTFOLIO	10000	5220.6	160.59	162.76	160.01

Conclusions

- SEDR simultaneously achieves **minimum Tweedie Deviance, maximum Gini Index, and near-zero ZMCE** — a three-objective dominance no competing strategy achieves.
- **Aleatoric querying** destroys rank-ordering by oversampling the **severity tail**; **Unbounded epistemic querying** destroys it by oversampling the **noise boundary**;
- Random querying preserves population structure without labeling leverage.

True Ensemble Epistemic Sensor ($K = 10$) ensemble utilizing algorithm-level stochasticity (feature subsampling, global seed dispersal, internal row subsampling without replacement) to generate mathematically pure **epistemic uncertainty scores** that are **decoupled from aleatoric claim severity**.

It demonstrated global optimum for this active learning pricing architecture, providing an industry-ready framework for pure premium/payout pricing in cold-start and low-data insurance market scenarios.

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